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Dynamic motion of red blood cells in simple shear flow

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A three-dimensional numerical model is proposed to simulate the dynamic motion of red blood cells (RBCs) in simple shear flow. The RBCs are approximated by ghost cells consisting of Newtonian liquid drops enclosed by Skalak membranes which take into account the membrane shear elasticity and the membrane area incompressibility. The RBCs have an initially biconcave discoid resting shape, and the internal liquid is assumed to have the same physical properties as the matrix fluid. The simulation is based on a hybrid method, in which the immersed boundary concept is introduced into the framework of the lattice Boltzmann method, and a finite element model is incorporated to obtain the forces acting on the nodes of the cell membrane which is discretized into flat triangular elements. The dynamic motion of RBCs is investigated in simple shear flow under a broad range of shear rates. At large shear rates, the cells are found to carry out a swinging motion, in which periodic inclination oscillation and shape deformation superimpose on the membrane tank treading motion. With the shear rate decreasing, the swinging amplitude of the cell increases, and finally triggers a transition to tumbling motion. This is the first direct numerical simulation that predicts both the swinging motion of the RBCs and the shear rate induced transition, which have been observed in a recent experiment. It is also found that as the mode changes from swinging to tumbling, the apparent viscosity of the suspension increases monotonically. © 2008 American Institute of Physics.

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I. INTRODUCTION

The dynamic motion of a liquid capsule enclosed by a thin elastic or incompressible membrane in shear flow has been studied experimentally,^{1–8} theoretically,^{9–13} and numerically^{14–23} by many researchers. The interest is mainly motivated by its relevance to blood flow in which blood cells and platelets are carried by plasma. Particular attention has been focused on the deformation of red blood cells (RBCs), which have a relatively greater effect on the rheology and hydrodynamic properties of blood.

RBCs are nucleus-free capsules. The membrane of a RBC consists of a lipid bilayer that is supported by a network of cytoskeletal proteins. The bilayer gives the membrane its nearly area-incompressible property while the cytoskeleton gives the elastic behavior. Thus the RBCs readily deform under shear forces while maintaining its surface area unchanged. The liquid inside RBCs is an aqueous solution of hemoglobin which is treated as Newtonian liquid. Without external force, a healthy human RBC has the biconcave discoid shape.

Much experimental work has been conducted to study the motion of RBCs under shear flow. Two types of motion are well known: a steady tank treading mode, in which a RBC deforms to a steady configuration with a fixed inclination and then the membrane rotates around the liquid inside; an unsteady tumbling mode, in which a RBC tumbles con-

tinuously. Recently, Abkarian *et al.*⁷ explored the motion of RBCs using a recent method of cell imaging parallel to the shear flow. They observed a novel “swinging” mode, which was said to happen between the two well-known modes. In the new mode, with the membrane rotating around the liquid inside, the RBC undergoes shape deformation and inclination oscillation. The swinging mode is actually an unsteady tank treading mode.

Several factors have been found to be important in determining a RBC’s behavior in shear flow. Goldsmith and Marlow² experimentally found that RBCs immersed in a low viscosity fluid tumble continuously, and RBCs immersed in a fluid with sufficiently high viscosity carried out tank treading motion. Keller and Skalak¹¹ theoretically analyzed the deformation of an ellipsoidal capsule in simple shear flow. It was found that for a capsule with a given geometry, the transition from tank treading mode to tumbling mode depends on the viscosity ratio between internal fluid and external fluid and it is independent of shear rate. The viscosity ratio dependent transition has also been studied and explained in subsequent works.^{12,15,20} Noguchi and Gompper^{21,22} studied vesicles with viscous membrane and found that increasing the viscosity of membrane can also cause the mode transition.

In the recent experiment of Abkarian *et al.*,⁷ besides the swinging motion it was also found that the swinging-to-tumbling transition of RBCs can be triggered by decreasing the shear rate. This shear rate induced transition of RBCs has also been mentioned by Fischer.⁸ These novel experimental findings cannot be recovered by the well-known theory of

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Keller and Skalak,¹¹ in which the capsule was assumed to have a fixed ellipsoidal shape with moving membrane. Based on the fixed-shape model of Keller and Skalak,¹¹ and under the further assumption that the membrane elastic energy undergoes a periodic variation during the tank treading motion, the experimental findings of Abkarian *et al.* can be successfully predicted by the theoretical model of Skotheim and Secomb,¹³ as well as Abkarian *et al.*⁷ In the recent numerical study of Noguchi and Gompper²³ on vesicles with viscous membranes, it was found that there is a shear rate induced transition of vesicles' motion from swinging mode to tumbling mode. Note, however, that the membrane of vesicles does not have shear elasticity, unlike that of RBCs. The free-energy landscape of a RBC is different from that of a fluid vesicle.²³ The membrane shear elasticity of RBCs prevents very large shape transformations,²⁴ such as a prolate-oblate transition observed for the fluid vesicles by Noguchi and Gompper.²³

In previous experiments on RBCs in simple shear flow, only two-dimensional (2D) images of the cells could be observed. The three-dimensional (3D) motion of the cells is not very clear. Another unclear issue is that in the tank treading region, how does a RBC change from the steady tank treading mode to the unsteady swinging mode. Direct numerical simulations have been attempted as an alternative way to address these questions. For initially biconcave discoid capsules with elastic membranes, it is expected that they will carry out the tank treading motion in simple shear flow if the internal liquid is the same as that outside and the shear rate is large.¹⁵ However, due to the complex geometry of the capsule, numerical instabilities were encountered in previous computations and the behavior was followed for an initial short stage.^{15–17} So far, the tank treading motion for the 3D biconcave capsules has not been fully recovered by direct numerical simulation, and neither has the shear rate induced transition of the capsules' motion from tank treading to tumbling been predicted.

In the present study, a numerical model is proposed and the dynamic motion of 3D RBCs in shear flow is investigated by direct numerical simulation. The RBCs are modeled as enclosed elastic and nearly incompressible membranes filled with Newtonian liquid which is the same as that outside. The cells have an initially biconcave discoid shape, which resembles the shape of a human RBC at rest. The capsule membrane is assumed to be infinitely thin and the bending stiffness is neglected.

The present study is based on the hybrid method of Feng and Michaelides,²⁵ as well as Sui *et al.*^{26,27} The immersed boundary concept^{28,29} was introduced into the framework of the lattice Boltzmann method.^{30,31} The flow field was solved by the lattice Boltzmann method, and the fluid-capsule interaction was solved by the immersed boundary method. The capsule membrane is discretized into flat triangular elements. A finite element model^{32,33} is incorporated to obtain the forces acting on the membrane nodes. The accuracy of the present model has been tested in a recent paper²⁷ by studying the transient deformation of spherical and oblate spheroidal capsules with various types of membranes in shear flow. The numerical results, including the Taylor shape parameter and

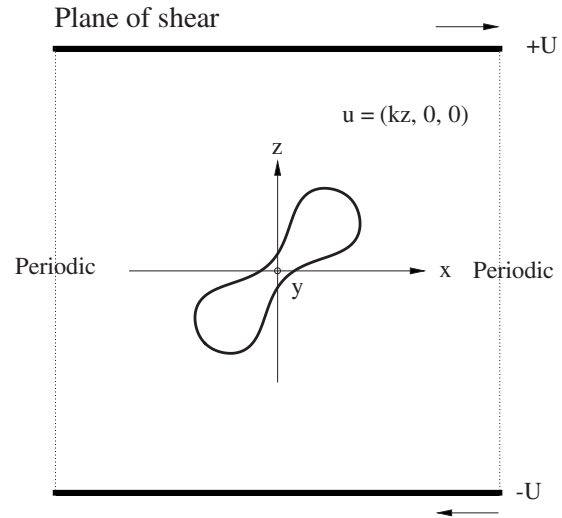


FIG. 1. A periodic suspension of initially biconcave discoid cells in shear flow.

the capsule inclination angle, were compared to previously published theoretical and numerical results. It was observed that for capsules under small deformation, the differences between the present numerical results and that of Barthès-Biesel⁹ by second order small deformation theory are within 2%; under large deformation, the differences between the present results and that of Lac *et al.*¹⁸ by the boundary element method are within 5%.

II. PROBLEM STATEMENT AND MEMBRANE MODEL

In the present study, the dynamic motion of a periodic file of 3D RBCs is considered to be subjected to the incident shear flow along the x -axis, $\mathbf{u} = (kz, 0, 0)$, as illustrated in Fig. 1. The shear flow is generated by two solid walls moving in opposite directions. Periodic boundary conditions are employed on the longitudinal direction and the spanwise direction. The membranes of the cells are very thin, and the bending stiffness of the membranes is neglected.

The constitutive law proposed by Skalak *et al.*³⁴ to model the membrane of RBCs is adopted in the present study. The strain energy function has the form

$$W = \frac{1}{4}E\left(\frac{1}{2}I_1^2 + I_1 - I_2\right) + \frac{1}{8}CEI_2^2, \quad (1)$$

where the terms I_1 and I_2 are the first and second strain invariants, with $I_1 = \lambda_1^2 + \lambda_2^2 - 2$ and $I_2 = (\lambda_1 \lambda_2)^2 - 1$. The terms λ_1 and λ_2 are the principle strains. In the right-hand side of the equation, the first term represents the shear effects, and second term accounts for the area dilation, with associated moduli E and CE , respectively. The term C is quite large for incompressible biological membranes.

In the present study, the 3D capsule membrane is discretized into flat triangular elements, following the approach of Ramanujan and Pozrikidis.¹⁵ To discretize the unstressed interface, each triangular face of a regular octahedron or icosahedron is subdivided into 4^n triangular elements. These elements are then projected radially onto a sphere. The geometry of each element is described by its three vertices. For

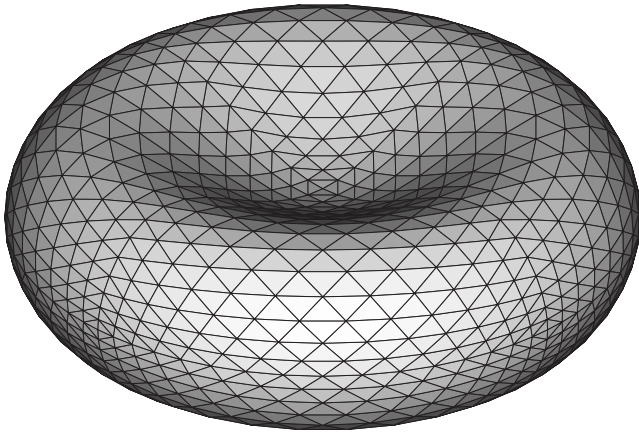


FIG. 2. Illustration of the interfacial discretization of a biconcave discoid cell.

the biconcave discoid, which is chosen to be the shape of RBC at rest, the mapping system is as follows:

$$\begin{aligned} x_{rbc} &= Rx, \quad y_{rbc} = Ry, \\ z_{rbc} &= 0.5R(1 - r^2)^{0.5}(0.207 + 2.003r^2 - 1.123r^4) \end{aligned} \quad (2)$$

where $r^2 = x^2 + z^2$ and the term R in Eq. (2) is the adjusting factor to keep the capsule volume constant. The discretization of a biconcave discoid surface is shown in Fig. 2.

The finite element membrane model developed by Charrier *et al.*³² as well as Shrivastava and Tang³³ is employed to obtain the forces acting at the discrete nodes of the membrane. In the finite element model, the membrane of the capsule is represented by a patchwork of flat triangular elements which remain flat after deformation, and the load on an element is represented by concentrated loads on the membrane nodes. Only in-plane stresses and strains exist. In the 3D deformation of the capsule, the membrane elements do not stay in the same plane after being displaced. In this model, the deformed element is transformed to the plane of the undeformed element so that the relative displacement of the nodes and the corresponding forces are easily determined. The equations developed for the force exerted at the nodes of an element of the deformed membrane are given in the plane of the element. The displacements are approximated by linear shape functions in the plane of the element with coordinates (x', y') . They are described by $\chi(x', y') = ax' + by' + c$. The unknown coefficients a , b , and c are determined using known values of the displacements at the three nodes of the element. After the displacements of the three nodes of an element are known, its states of strain λ_1 and λ_2 can be obtained. The relation of membrane strain and membrane energy is governed by the constitutive law, which is determined by the membrane materials. With a constitutive law chosen, one can follow the usual finite element procedure and derive the relations between nodal forces and nodal displacements. The principle of virtual work is used to calculate the forces at the three nodes of an element, which are finally in the form of

$$\begin{aligned} \{F_x\} &= V_e \frac{\partial W}{\partial \lambda_1} \left\{ \frac{\partial \lambda_1}{\partial u} \right\} + V_e \frac{\partial W}{\partial \lambda_2} \left\{ \frac{\partial \lambda_2}{\partial u} \right\}, \\ \{F_x\} &= V_e \frac{\partial W}{\partial \lambda_1} \left\{ \frac{\partial \lambda_1}{\partial v} \right\} + V_e \frac{\partial W}{\partial \lambda_2} \left\{ \frac{\partial \lambda_2}{\partial v} \right\}, \end{aligned} \quad (3)$$

where $\{F_x\}$ and $\{F_y\}$ are the nodal forces in the x and y directions corresponding to the current deformed state of the membrane element, V_e is the original area of the element, W is the strain energy density, and u and v represent the nodal displacements. Because each node of the discrete membrane belongs to more than one element, the resultant force on a node is the sum of the forces exerted by the m elements attached to the node. So far, the force calculated is the fluid force acting on the capsule membrane. Its equal and opposite counterpart is the force acting on the fluid. It is distributed to the surrounding fluid by the approach described by the immersed boundary method. The above is only a brief description of this model, and the detailed treatment can be found in pertinent literatures.^{32,33}

III. NUMERICAL METHOD

The present study is based on a hybrid method,^{25–27} which is a combination of the immersed boundary method and the lattice Boltzmann method. In the immersed boundary method of Peskin,^{28,29} a force density is distributed to the Cartesian mesh in the vicinity of the moving boundary in order to account for the effect of the boundary. Two different sets of coordinate system are used. The fluid region is represented by Eulerian coordinates $\mathbf{x}$ and the moving boundary immersed in the fluid are represented by Lagrangian coordinates $\mathbf{s}$. Any position on the capsule membrane can be written as $\mathbf{x} = \mathbf{X}(\mathbf{s}, t)$. The term $\mathbf{F}(\mathbf{s}, t)$ represents the membrane force density, which is a combination of the internal link force induced by deformation and the external force. The term $\mathbf{f}(\mathbf{s}, t)$ represents the fluid body force density.

The nonslip boundary condition is satisfied by letting the flexible membrane move at the same velocity as the fluid around it. That is,

$$\frac{\partial \mathbf{X}(\mathbf{s}, t)}{\partial t} = \mathbf{u}[\mathbf{X}(\mathbf{s}, t), t]. \quad (4)$$

This motion will cause the capsule to deform. The boundary force density $\mathbf{F}(\mathbf{s}, t)$ is obtained from the constitutive law of the elastic membrane and distributed to the fluid mesh points near it by a smoothed version of the Dirac delta function, written as follows:

$$\mathbf{f}(\mathbf{x}, t) = \int_{\Gamma} \mathbf{F}(\mathbf{s}, t) \delta[\mathbf{x} - \mathbf{X}(\mathbf{s}, t)] d\mathbf{s}, \quad (5)$$

where

$$\delta(r) = \begin{cases} \frac{1}{4} \left[1 + \cos\left(\frac{\pi|r|}{2}\right) \right], & r \leq 2 \\ 0, & r > 2, \end{cases} \quad (6)$$

in which r equals the distance between the Lagrange and Eulerian nodes divided by the Eulerian grid space. For 3D

case, the smoothed version of the Dirac delta function can be written as

$$\delta(\mathbf{X} - \mathbf{X}_{ij}) = \delta(x - x_{ij})\delta(y - y_{ij})\delta(z - z_{ij}). \quad (7)$$

The same smoothed version of the Dirac delta function is used to obtain the velocities of the Lagrangian nodes on the moving boundary.

The lattice Boltzmann method is a kinetic-based approach for simulating fluid flows. It decomposes the continuous fluid flow into pockets of fluid particles. In the present study, the immersed boundary method is combined with the lattice Boltzmann method. In order to solve the flow field with a force density, the lattice Boltzmann equation must be modified. Several forms of LBE which can handle a force density have been proposed. The approach of Guo *et al.*³⁵ is

more accurate for unsteady flow with the force changing with time and space in which the modified lattice Boltzmann equation is in the form of

$$f_i(\mathbf{x} + \mathbf{e}_i \Delta t, t + \Delta t) - f_i(\mathbf{x}, t) = -\frac{1}{\tau} [f_i(\mathbf{x}, t) - f_i^{\text{eq}}(\mathbf{x}, t)] + \Delta t F_i, \quad (8)$$

where $f_i(\mathbf{x}, t)$ is the distribution function for particles with velocity $\mathbf{e}_i$ at position $\mathbf{x}$ and time t , Δt is the lattice time interval, $f_i^{\text{eq}}(\mathbf{x}, t)$ is the equilibrium distribution function, and τ is the nondimensional relaxation time.

The D3Q19 model is one of the most commonly used LB models for 3D simulation. The fluid particles have possible discrete velocities stated as follows:

$$[\mathbf{e}_0, \mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3, \mathbf{e}_4, \mathbf{e}_5, \mathbf{e}_6, \mathbf{e}_7, \mathbf{e}_8, \mathbf{e}_9, \mathbf{e}_{10}, \mathbf{e}_{11}, \mathbf{e}_{12}, \mathbf{e}_{13}, \mathbf{e}_{14}, \mathbf{e}_{15}, \mathbf{e}_{16}, \mathbf{e}_{17}, \mathbf{e}_{18}] = \begin{bmatrix} 0 & 1 & -1 & 0 & 0 & 0 & 0 & 1 & 1 & -1 & -1 & 1 & -1 & 1 & -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & -1 & 0 & 0 & 1 & -1 & 1 & -1 & 0 & 0 & 0 & 0 & 1 & 1 & -1 & -1 \\ 0 & 0 & 0 & 0 & 0 & 1 & -1 & 0 & 0 & 0 & 0 & 1 & 1 & -1 & -1 & 1 & -1 & 1 & -1 \end{bmatrix}. \quad (9)$$

The equilibrium distribution function $f_i^{\text{eq}}(\mathbf{x}, t)$ is in the form of

$$f_i^{\text{eq}} = E_i(\rho, \mathbf{u}) \quad (10)$$

with

$$E_i(\rho, \mathbf{u}) = \omega_i \rho \left[1 + \frac{\mathbf{e}_i \cdot \mathbf{u}}{c_s^2} + \frac{\mathbf{u} \mathbf{u} : (\mathbf{e}_i \mathbf{e}_i - c_s^2 \mathbf{I})}{2c_s^4} \right], \quad (11)$$

where ω_i is the weighing factor, it equals 1/3 for $i=0$, 1/18 for $i=1-6$, and 1/36 for $i=7-18$. The term c_s represents the sound speed and equals $\Delta x / (\sqrt{3} \Delta t)$.

The relaxation time is related to the kinematic viscosity in Navier–Stokes equation in the form of

$$\nu = \left(\tau - \frac{1}{2} \right) c_s^2 \Delta t. \quad (12)$$

The forcing term F_i is in the form of

$$F_i = \left(1 - \frac{1}{2\tau} \right) \omega_i \left[\frac{\mathbf{e}_i - \mathbf{u}}{c_s^2} + \frac{(\mathbf{e}_i \cdot \mathbf{u})}{c_s^4} \mathbf{e}_i \right] \cdot \mathbf{f}. \quad (13)$$

Once the particle density distribution is known, the fluid density and momentum are calculated using

$$\rho = \sum_i f_i, \quad \rho \mathbf{u} = \sum_i \mathbf{e}_i f_i + \frac{1}{2} \mathbf{f} \Delta t. \quad (14)$$

IV. RESULTS AND DISCUSSION

In the present study, the length scale, velocity scale, and time scale are chosen to be a , ka , and $1/k$, respectively. The term a is the equivalent radius, defined as $a = (3V/4\pi)^{1/3}$, where V is cell volume. The term k is the shear rate. The

Reynolds number based on the length and velocity scales is 0.1 so that Stokes flow condition is satisfied and inertia effect is negligible. The dimensionless shear rate is identified to play an important role in determining the cell's deformation. The dimensionless shear rate is defined as $G = \mu ka / E$, where μ is the viscosity of the surrounding fluid. The shear elasticity modulus is adjusted to achieve different dimensionless shear rates.

The linear shear flow is generated by two solid walls, with distance $5a$ apart, moving in opposite directions. The computational domain is a cuboid box with side length $6a$ in the x -axis, and $5a$ in the y - and z -axis. The cell is at the center of the domain. The boundary conditions at the upper and lower planes of the computational domain are set as solid walls moving in opposite directions. Periodic boundary conditions are employed on the longitudinal and spanwise directions. The grid resolutions are $\Delta x = \Delta y = \Delta z = a/16$. The membrane is discretized into 32 768 flat triangular elements connecting 16 386 nodes. Grid convergence studies showed that the mesh resolutions are sufficient.

It is known that the RBC membrane has strong resistance to area dilatation, and in Skalak's law [Eq. (1)] the ratio between the membrane area dilatation modulus and shear elasticity modulus is on the order of 10^5 . Such a large ratio makes the present computation diverge. Eggleton and Popel¹⁶ observed similar instability and proposed the use of much smaller ratios. In order to keep the membrane area nearly unchanged, the moduli ratio should increase with the shear rate, but not so large to make the computation diverge. In the present study, the approach of Eggleton and Popel is adopted. With the dimensionless shear rate ranging from

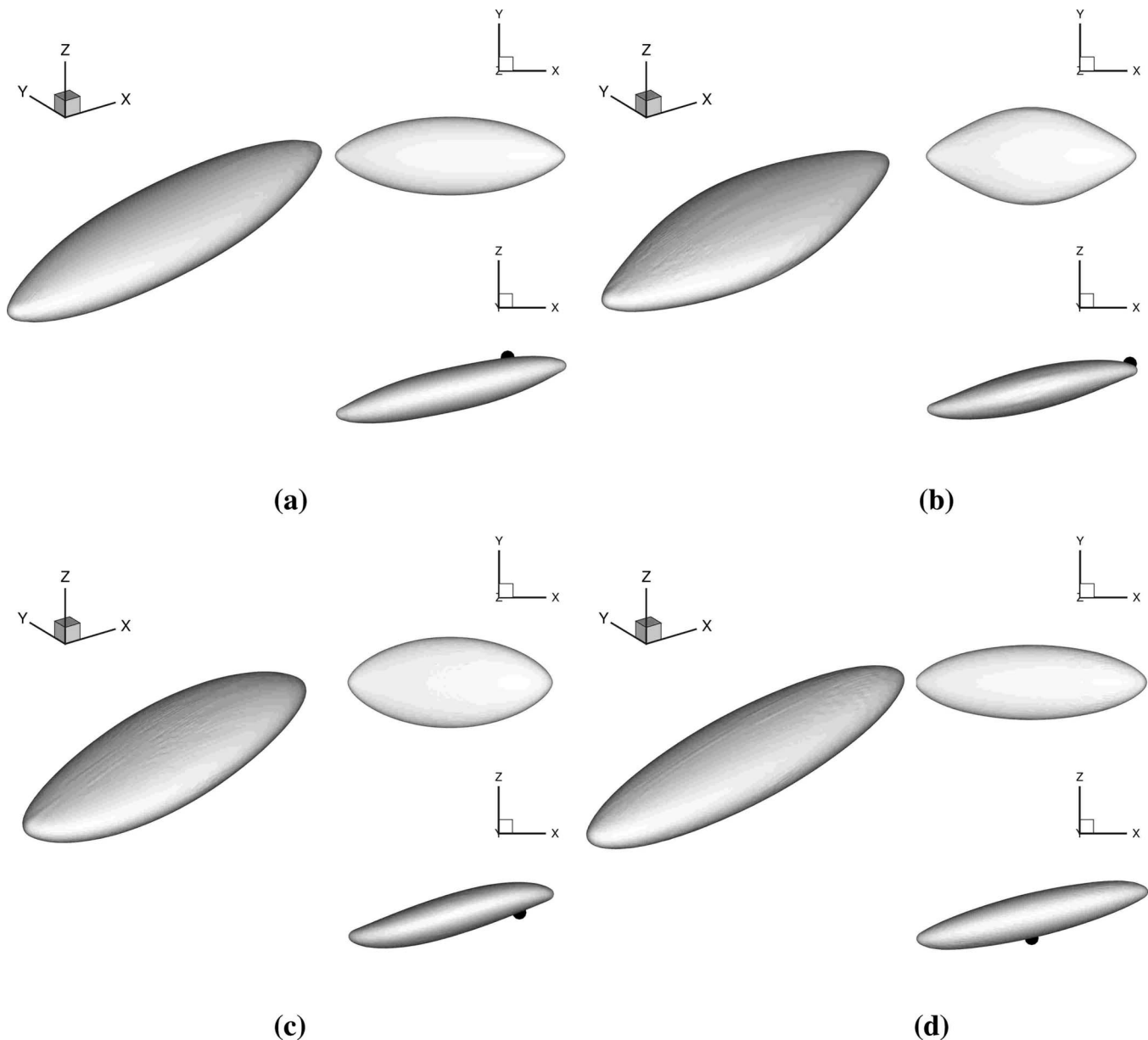


FIG. 3. Profiles of the cell in the swinging motion at $G=1.87$ and kt is equal to (a) 5, (b) 9, (c) 13, and (d) 17.

0.0005 to 2.8, the moduli ratio C ranges from 1 to 400. For all cases considered, both the membrane area changes and the cell volume changes are within 0.5%.

A. Swinging motion

Many experiments have been carried out to study the dynamic motion of RBCs in shear flow and different experimental techniques have been employed. Schmid-Schönbein and Wells,¹ Fischer *et al.*,³ as well as Tran-Son-Tay *et al.*⁴ used a counter-rotating cone plate to generate a shear chamber and observed the motion of RBCs from a 2D plane parallel to the rotating plate. The steady tank treading motion of RBCs at large shear rates was reported. Recently, Abkarian *et al.*⁷ studied the motion of RBCs using a new method of cell imaging, which gave 2D images of the cell in the plane of shear, and a novel swinging mode was observed at moderate shear rate. It would be quite interesting to know how a

cell transits from the steady tank treading mode to the unsteady swinging mode with the shear rate decreasing.

In the present study, first considered is the dynamic motion of RBCs in simple shear flow at moderate and large shear rates. From the simulation results, it is found that the cell deforms to oblate ellipsoidal shapes and then the membrane rotates around the internal liquid. During this tank treading motion, the cell inclination oscillates periodically and the cell profile undergoes periodic vacillating-breathing deformation. These behaviors are consistent with the swinging mode reported by Abkarian *et al.*⁷ The transient 3D and 2D profiles of the cell during the swinging motion at $G=1.87$ are presented in Fig. 3. In the 2D images of the cell in x - z plane, which is parallel to the shear plane, the black dots represent the same material point on the cell membrane. From the 2D images, it can be observed that the cell inclination is undergoing small amplitude variation while the cell

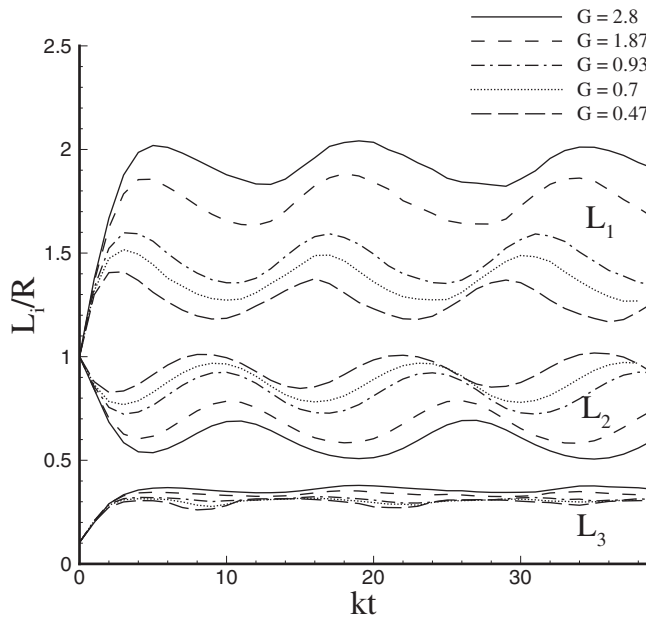


FIG. 4. Temporal evolution of the cell's length (L_1), width (L_2), and thickness (L_3) under various dimensionless shear rates.

membrane is rotating around the liquid inside. It can also be observed that there is vacillating-breathing deformation of the cell, as observed from the 2D images in x - y plane, which is parallel to the solid moving wall, as well as from the 3D profiles of the cell.

The temporal evolutions of the cell's length, width, and thickness under various dimensionless shear rates are presented in Fig. 4. In the present study, the cell's length and thickness are defined as the maximum length and breadth of the cell cross section in the plane of shear; and the cell width is defined as the maximum length of the cell in the spanwise direction. The cell inclination is defined as the inclination of the cell length axis. The temporal evolutions of the cell inclination are presented in Fig. 5. At larger shear rate, the cell is more elongated and more aligned with the flow direction. With the shear rate ranges from 0.47 to 2.8, the cell inclination oscillation amplitude is in the range of about $2.9^\circ - 3.5^\circ$. The oscillation amplitudes of the cell inclination under various dimensionless shear rates are plotted in Fig. 6. It is seen that with the shear rate increasing, the amplitude decreases toward zero. It is thus concluded that the steady tank treading mode is absent if the shear rate is not infinitely high.

From the results by direct numerical simulation, the steady tank treading motion has been only observed for initially spherical capsules.^{14,15,18} In the numerical study of Ramanujan and Pozrikidis¹⁵ of initially nonspherical capsules with elastic membranes in simple shear flow, swinging motion of the capsules, similar to the present study, was observed. These results show that in shear flow, for initially nonspherical capsules with membranes having shear elasticity, the steady tank treading mode is absent. This is because such capsules have shape memory effect, as had been experimentally observed for RBCs by Fischer.⁸ For spherical capsule, the membrane shear elastic energy can remain unchanged during steady tank treading motion because the

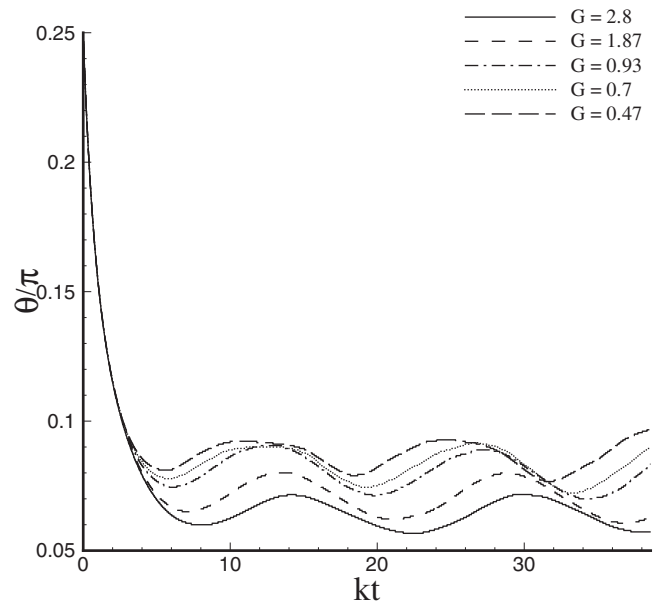


FIG. 5. Temporal evolution of the cell inclination under various dimensionless shear rates.

membrane material points are equivalent, and thus the steady mode can be maintained. However, for nonspherical capsules, during the tank treading motion, the membrane elastic energy undergoes periodic variation due to the shape memory effect. The energy variation will provide a perturbation force which prevents the motion from being steady. The initially nonspherical capsules thus carry out swinging motion in shear flow when the shear rate is high.

In the experimental study of Fischer *et al.*,³ steady tank treading motion of RBCs was reported, and 2D profiles of a RBC in the steady tank treading motion were presented in their Fig. 3(a). However, a closer examination shows that the cell was undergoing a slight vacillating-breathing deformation during the tank treading motion, which has some simi-

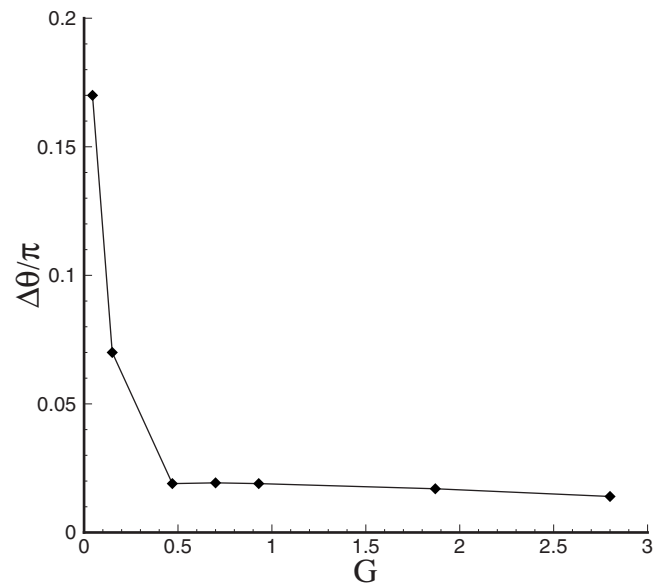


FIG. 6. Amplitude of the cell inclination oscillation under various dimensionless shear rates.

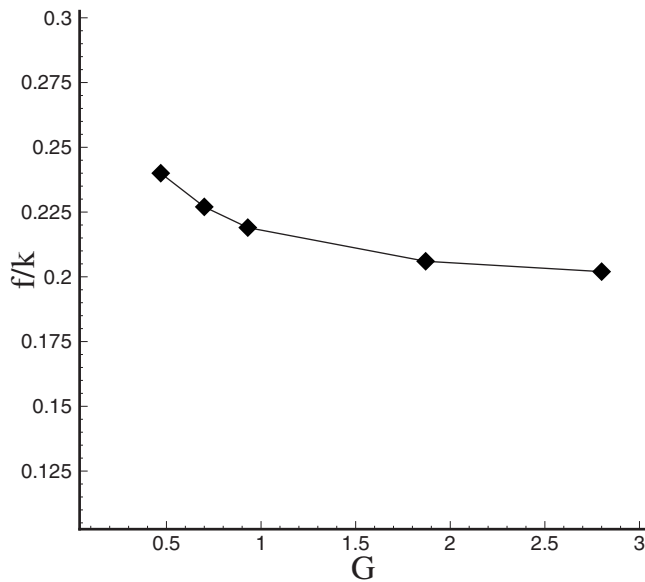


FIG. 7. Tank treading frequency of the cell membrane under various dimensionless shear rates.

larities to the results in the present Fig. 3. In the results of Fisher *et al.*, the maximum length of the cell varies with amplitude of about 5%. This may suggest that it was not in a steady mode. Also, from the theory of Skotheim and Secomb,¹³ with the capsule changing shape (or membrane elastic energy changing) during the tank treading motion, its inclination is unlikely to be fixed.

At very high shear rates, because the fluid shear stress is relatively larger compared with the membrane elastic tension, the swinging amplitude will be very small (inclination oscillation amplitude less than 3° in the present study for $G=2.8$). Without high definition imaging technique and suitable viewing plane, it could be rather difficult to discriminate small amplitude swinging motion from steady tank treading motion in experiments. This is especially so because both swinging motion and steady tank treading motion belong to the tank treading regime.

The time T for the cell membrane to complete a rotation is computed to obtain the tank treading frequency $f=2\pi/T$. Since the membrane has a complicated 3D motion, the tank treading period is taken to be the time for a membrane point, on the cell cross section in the plane of shear, to complete a rotation. The nondimensional frequency f/k is presented in Fig. 7. It is found that the normalized tank treading frequency is higher at lower dimensionless shear rate and ranges from 0.20 to 0.24 with the dimensionless shear rate ranging from 0.47 to 2.8. In the experiment of Tran-Son-Tay *et al.*⁴ it was found that for younger RBCs suspended in a medium of comparable viscosity, f/k equals 0.22. In relating the present estimates of the tank treading frequency with experiments, it should be noted that there is non-negligible wall confining effect. In the present study the walls were kept apart at only five equivalent RBC radius. This choice was made purely to avoid heavy computational load. The deformation of a vesicle had been found to be quantitatively influenced by a shallow channel,²⁰ and similar influence may be expected on RBCs. In most previous experiments, the

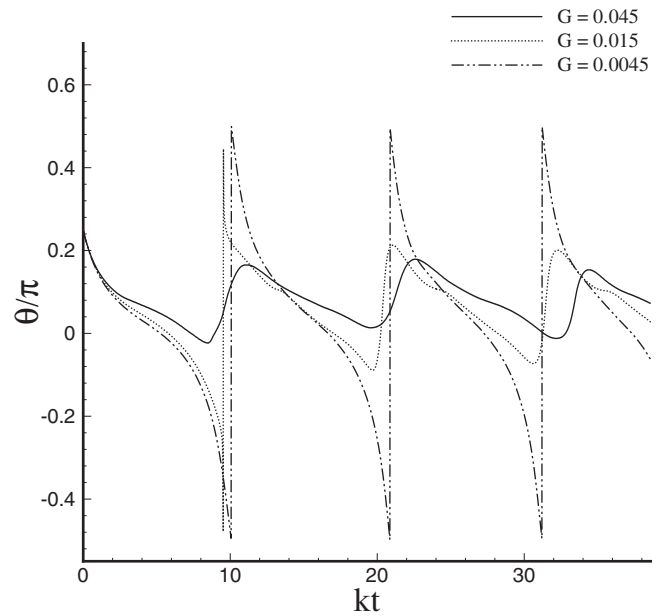


FIG. 8. Temporal evolution of the cell's inclination under various dimensionless shear rates.

wall distances were larger than that in the present simulation. Thus any quantitative comparison with previous experiments has to be viewed with some circumspection.

B. Swinging-to-tumbling transition

Swinging motion of a cell consists of periodic inclination oscillation and shape deformation superimposed on the tank treading motion. For RBCs, it has long been recognized that the tank treading to tumbling transition can be induced by increasing the internal liquid viscosity or membrane viscosity of the cell. In both cases, due to the increase of viscosity, the transfer of shear torque to the membrane becomes more and more difficult, and then the capsule would behave like a solid body which undergoes tumbling.

Recently, Abkarian *et al.*⁷ experimentally found that the swinging-to-tumbling transition of RBCs can be triggered by decreasing the shear rate. The phenomenon has also been mentioned by Fischer.⁸ Noguchi and Gompper²³ numerically predicted similar shear rate induced transition on vesicles. These novel findings advance the state of people's knowledge in this field.

The dynamic motion of cells in simple shear flow at small dimensionless shear rates is considered in the present study. Figure 8 presents the temporal evolutions of the cell's inclination at $G=0.0045$, 0.015 , and 0.045 . It is seen that at $G=0.045$ the cell still carries out swinging motion with amplitude much larger than that presented in Fig. 5 due to the lower shear rate.

At $G=0.015$, it is seen that the temporal evolution of the cell's inclination shows a discontinuity at $kt=9.5$, and the oscillation amplitude equals π , which is an indication of tumbling. After the initial tumbling, the cell carries out swinging motion. In the theory of Skotheim and Secomb,¹³ the swinging-to-tumbling transition of an elastic ellipsoidal capsule was found to occur through a narrow intermittent

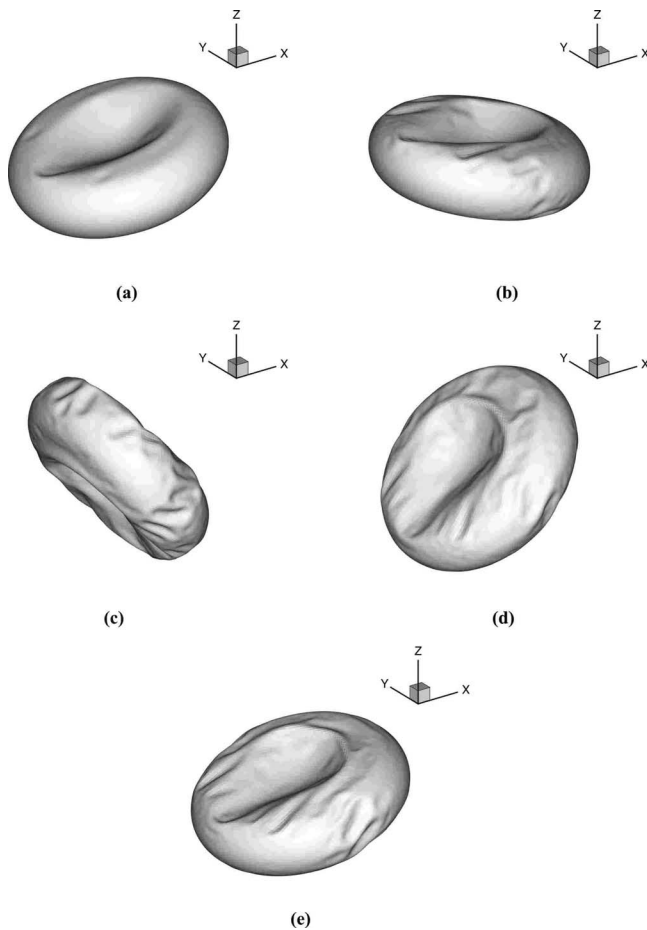


FIG. 9. 3D profiles of the cell during the flipping motion. The dimensionless time kt is equal to (a) 2, (b) 6, (c) 9, (d) 11, (e) 13, and $G=0.0005$.

regime of successive swinging and tumbling. Experimental evidence has also been reported by Abkarian *et al.*⁷ The present simulation may suggest a similar intermittent motion of the cells; however, there is insufficient evidence because in much longer computation of the present study, the membrane wrinkles, due to a lack of bending resistance, became severe and the computation results tended to become unreliable.

With further decrease in the shear rate, the cell's motion goes into the tumbling region, as shown in Fig. 8 for $G=0.0045$, the oscillation amplitude of the cell inclination equals π . The 3D profiles of the cell during the tumbling motion at $G=0.0005$ are presented in Fig. 9. It can be seen that slight wrinkles develop on the cell interface.

The membrane wrinkles are not observable during the present simulation for small amplitude swinging motion at large shear rates, as shown in Fig. 3; however, the wrinkles become visible during the transition and the tumbling motion. Membrane wrinkles have been first experimentally observed by Walter *et al.*⁶ on polymerized capsules in shear flow. As explained by Lac *et al.*,¹⁸ the wrinkles indicate that the membrane develops compressive stress locally due to a lack of sufficient bending rigidity. Subsequently, Lac and Barthès-Biesel¹⁹ proposed the imposition of an isotropic tensile prestress on the membrane to avoid wrinkling. In the present study, the simulation accuracy has been affected by

the neglecting of the membrane bending rigidity, the inclusion of which is recommended in future work in order to reduce the membrane wrinkling. In addition, numerical smoothing of the capsule's surface may be necessary, as reported in the numerical study of Pozrikids¹⁷ on the tank treading and tumbling motion of highly flatten capsules, for example, RBCs.

The recent experimental findings of Abkarian *et al.*⁷ for RBCs, as well as the present numerical results for modeled RBCs, show that in shear flow, the swinging-to-tumbling transition can be induced by lowering the shear rate. This is different from the theoretical prediction of Keller and Skalak,¹¹ which showed that the transition was only related to the viscosity ratio. The shear rate induced transition is attributed to the shape memory effect of the 3D capsules.

In Keller and Skalak's theory, the capsule was assumed to have a fixed ellipsoidal shape during the membrane tank treading motion, and the membrane elastic energy was thus kept unchanged. These assumptions have excluded the shape memory effect. Recently, based on the fixed-shape model of Keller and Skalak, Skotheim and Secomb¹³ included an additional membrane elastic energy variation during the tank treading motion (which actually includes the shape memory effect), both the swinging motion and shear rate induced swinging-to-tumbling transition can be successfully predicted. Different from the previous theoretical approach, the present work is by direct numerical simulation which includes cell deformation.

Lowering the shear rate could induce the swinging-to-tumbling transition, this happens when the shear rate is very low, the fluid shear force is relatively small and cannot overcome the membrane elastic energy gradient during the tank treading motion. The cell is then relatively solidified and can only tumble like a rigid body.

C. Rheological property

The drag force F_D exerted by the fluid on the upper or lower wall is studied under various dimensionless shear rates. The drag coefficient is defined as

$$C_D = \frac{F_D}{\mu \frac{2U}{5a}(5a6a)} = \frac{F_D}{12\mu Ua}. \quad (15)$$

The drag coefficient may be regarded a measure of the effective shear viscosity of the suspension. Figure 10 presents the temporal evolutions of the drag coefficient at various dimensionless shear rates. Shear thinning effect is clearly observed, in which the average shear viscosity of the suspension is higher at lower dimensionless shear rates.

It must be noted that in the present simulation, the viscosity ratio between the internal liquid of the RBC and the external liquid is assumed to be unity as was also used in some previous studies.^{16,36} For healthy RBCs suspended in plasma, the viscosity ratio of about 5 is generally accepted. It is well known that the viscosity contrast between the internal and external liquids has an important role on the type of motion of a RBC. A RBC with a higher internal viscosity is more liable to tumble for a given shear rate. The assumption

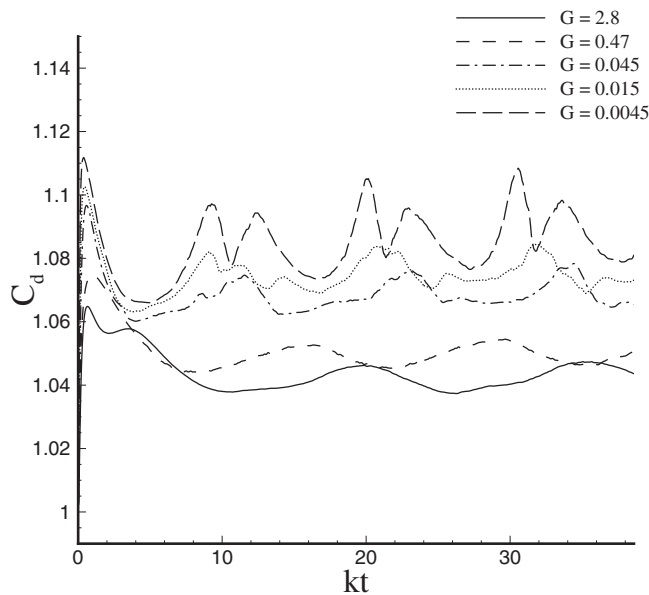


FIG. 10. Temporal evolutions of the drag coefficient of the upper wall under various dimensionless shear rates.

of unity viscosity ratio in the present study is mainly due to the restriction of the presently used immersed boundary method. This restriction may be relaxed by employing the front-tracking method,^{37–39} which, however, is presently restricted to 2D cases.^{38,39}

V. CONCLUSION

Based on a hybrid model, the dynamic behavior of initially biconcave discoid capsules with Skalak membranes is studied under shear flow to mimic the motion of RBCs. At large shear rate, the cells carry out a swinging motion in which periodic inclination oscillation and shape deformation superimposes on the membrane tank treading motion. With the shear rate decreasing, the swinging amplitude of the cell increases, and finally triggers a transition to tumbling motion. Both the swinging motion of the initially biconcave cell and the transition induced by lowering the shear rate agree with recent experimental findings on RBCs, and may be due to the shape memory effect of the cell. From swinging to tumbling, the apparent viscosity of the suspension is found to decrease monotonically.

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